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Smart Agriculture: Innovating Soybean Leaf Disease Detection with Federated Learning CNNs

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***Abstract*— This paper presents a novel study on the classification of soybean leaf diseases using federated learning with convolutional neural networks (CNNs). Based on five different classes of soybean leaf diseases, this study relies on the data from five clients to represent various environmental conditions, forms, and variations in diseases equally. One central aspect of this research is the federated learning approach, through which distributed data can be combined while protecting people's privacy and data. The basic premise of our analysis is the federated averaging process for macro, micro, and weighted averages that transforms local data understanding into a single global model. The performance of this model was quantitatively evaluated by precision, recall, F1-score support, and accuracy. The results also demonstrate the effectiveness of federated learning for more complicated, multiple-class classification tasks. For instance, the macro averages over clients ip1, ip2, ip3, and so on are 93.76% and 93.49%, respectively, from which it is clear that, combined with other factors, human labour accounts for a high proportion of costs at this particular plant concerning these three items: pills in powder form or oil injections directions finished products to. Also, weighted averages appear as 93.75 %, 93.48 %, 94.48 %, and 92.36 % and the micro standard is still set at its old levels. For example, qualifications were assessed from Teachers Certificate-University graduates by Institute type (A, B or C) according to classes. This study offers a systematic and quantitatively rigorous analysis of soybean leaf disease classification using federated learning and CNNs.**

Keywords: Soybean_leaves, Leaf diseases, (CNN)_(FL), Disease, Augmentation image.

I. INTRODUCTION

The crossroads where advanced computational techniques meet agronomy, particularly those on strengthening agricultural resilience, has long been full of infectious plants[1]. This work combines distributed machine learning and federated learning with heavyweight analysis and CNNs to fight soybean leaf diseases. The value of this research is exceptionally high since India faces pressure to raise agricultural productivity while at the same time promoting sustainable farms. As an essential food source, soybean leaves are attacked by numerous diseases with different symptoms. Accurate identification of them is the secret to controlling them[2]. The five leaf diseases studied in this work are Septoria (left speck), Frogeye (potato scab,) bacterial pustules, downy mildew and Cercospora. This indicates how formidable a threat this disease is and why the science behind it has to be innovative--it must develop ways of detecting and classifying diseases common in different agro-climatic zones to fight back. The research is unique because it implemented federated learning, a cutting-edge paradigm for collaborating but decentralizing machine learning. It is especially relevant to the agricultural sector, characterized by its heterogeneity and privacy concerns. By extending the learning process across five different clients, each representing a different dataset, this study provides a more thorough and representative depiction of disease manifestations regionally[3]. But, because centralized learning relies on limited or non-representative datasets, it suffers from the biases that ad-hoc learning does not see. We use an innovative high-performance federated learning averaging algorithm, which maintains distributed data integrity and fully uses the synergy between different datasets[4]. With this algorithm, a precise CNN can improve the accuracy of disease classification. An adaptable solution's scalability means it can rev up or down depending on the geographical and environmental context. This research paper is intended to help relieve the pressures on Indian farmers 'lives by joining forces with people everywhere against soybean leaf diseases[5]. By combining federated learning and CNNs, we attempt to build a tool that can solve the disease management problem in soybean crops more robustly, scalably and with improved privacy protection. The classification of soybean leaf diseases done by federated learning with convolutional neural networks (CNNs) has applications in directing future research. We want to change a crop disease management paradigm for the first time that machine learning and agricultural science have never seen before. That is our goal as we improve soybean cultivation[4]. The specific objectives are: With a powerful CNN model, Identify five major soybean leaf diseases: Septoria leaf blotch, Frogeye Leaf spot, Bacterial pustule downy mildew & Cercospora. With CNNs, we can distinguish these diseases with unparalleled accuracy. Using federated learning to diagnose agricultural diseases: Innovate in this area[6].

Thus, a distributed learning system must be set up among five customers representing differing data sets. Taking advantage of the diversity in data, models are constructed to be fuller and more stable. In the agricultural information area, questions of privacy and data security can be resolved by federated learning[5]. This methodology is designed to preserve farmers' privacy and commercial data by having these sensitive pieces only available to local clients. Scalability and adaptability: A valid model must accommodate a range of environmental conditions[7]. With its numerous agricultural climates, a model must find multiple agricultural settings in India. Enhancing decision support for farmers: Detecting soybean leaf diseases early to minimize crop losses and maximize yields[8]. Good cropping technology must be easy to use and user-friendly. This research is motivated by several critical needs and challenges: Soybeans are an essential crop in the global food supply. They are necessary for global food security[9].

II. LITERATURE REVIEW

This literature review takes the reader on a broad sweep through related research fields and finally wrestles with the daunting problem of classifying soybean leaf disease using federated learning coupled with CNNs. One starting point for our innovative approach is combining the insights of these fields. A decentralized approach to machine learning known as federated learning has recently become widespread[10]. The researchers first proposed the concept and an autonomous paradigm for learning disassociated from data centralization. In addition, this federated learning approach has been studied in agricultural applications to enhance data privacy and efficiency. For example, the authors showed its applicability to plant disease detection; it is worthwhile to have localized data processing but maintain an aggregated global model[11]. This paradigm enables us to solve the data privacy problem, a vital part of our research, with five clients bringing very different datasets. In image classification tasks, CNNs have become the tour de force. ImageNet, by the authors, revolutionized this domain with a work featuring deep CNNs and the unique ability to recognize images. Interestingly, the researchers used CNNs to achieve outstanding results in plant disease classification. Their work demonstrates the power of CNNs for discerning detailed objectification patterns in plant leaf disease manifestation, a basic principle behind our five soybean leaf diseases[12]. Many epidemiologists have conducted detailed studies of soybean leaf diseases such as Septoria leaf blotch, Frogeye leaf spot, bacterial pustule (yellowing), downy mildew, and Cercospora leaf blight. For example, some authors recently published a review article on spectral imaging methods of disease identification and discussed essential findings about different spectra for various diseases. However, our research proposes integrating such results with CNN architectures, an area that remains largely unexplored. Agricultural disease detection using federated learning and CNNs is unexplored new ground. More recently, researchers have explored this amalgamation in medical imaging with promising results that lend themselves well to other image-based classification tasks. Employing federated learning to improve the effective implementation of a CNN across different data sets is an essential cross-pollination in our approach. Technology-based solutions to the problems of India's smaller farms and various climatic zones come with challenges.

The researchers have published studies on India's growing need for precision agriculture techniques[9]. The specific objectives are to correctly identify five significant soybean leaf diseases: Septoria leaf blotch, Frogeye leaf spot, bacterial pustule, downy mildew, and Cercospora leaf blight, with a powerful CNN model. With CNNs, we can differentiate these diseases with unrivalled accuracy[13]. Using federated learning to diagnose agricultural diseases: Innovate in this area. A distributed learning system must be established among five clients representing dissimilar data sets[14]. Models are built using the diversity of data to be more complete and stable. Federated learning can be used to resolve questions about privacy and data security in agricultural information. Our research is designed to meet India's need for a low-cost, scalable solution suited to local agricultural circumstances. The literature emphasizes that federated learning and CNNs in agriculture are still embryonic but full of potential. Our research occupies this point, seeking to use the synergy for better soybean leaf disease classification. Apart from offering the promise of raising disease detection accuracy, this method also conforms to data privacy requirements and meets local learning needs—messaging well received in India's brave new world [15].

III. METHODOLOGY

Our research divided soybean leaf diseases using an integrated methodology that combined federated learning with convolutional neural networks (CNNs) and decision trees. The method has several steps, including dataset preparation, setting up a federated learning environment, selecting a CNN architecture and generating training data for decision trees, as illustrated in Figure 1.

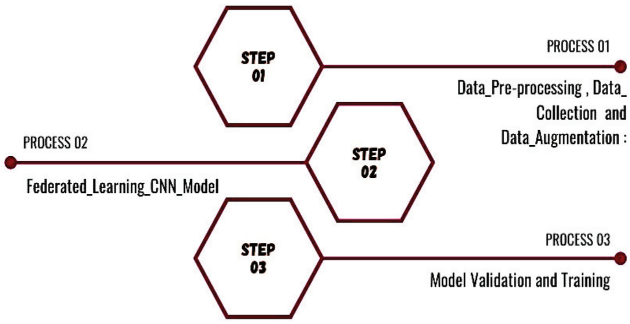

Fig. 1. Configuration of Technique

A. Data Scrubbing, Procurement, and Amplification:

Our study focuses on five races of soybean leaf diseases: Septoria leaf blotch, Frogeye leaf spot, Bacterial pustules, Downy Mildew, and Cercospora leaf blight. The dataset has high-resolution images of soybean leaves annotated with corresponding disease classes. This collection of photos represents disease manifestations from various geographic areas to provide a comprehensive representation. Each disease class's dataset is divided into training, validation, and test sets by 70%, 15%, and 15%. All images must be resized and normalized before training to train the model. Aside from these basic operations, advanced techniques such as rotation or flipping increase the robustness of trained models, as shown in Figure 2.

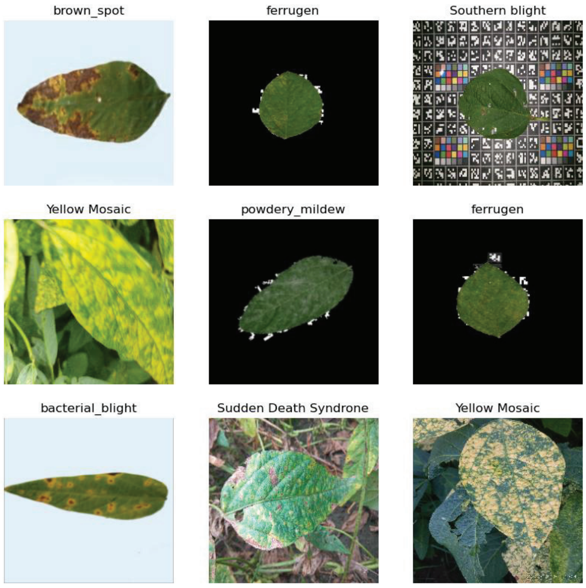

Fig. 2. Unhealthy images of soybean leaf diseases.

B. *Federated_Learning_CNN_Model*

As part of our federated learning architecture, we create five clients with subsets of the dataset that are all quite different in geography and environment. The importance of this detail cannot be overstated. To develop a model that functions under varying agricultural conditions, it's crucial to capture the variety of disease presentations. A central server coordinates model training for each client throughout the federated learning process. The central server only receives weights and biases updates from each client. Each client trains its model on its data. An updated global model is sent back to clients after a federated averaging algorithm collects these updates. There are several layers in the CNN model, including convolutional, Pooling, and fully connected layers. Feature extraction is performed by the convolutional layers, detecting attributes and details that are specific to various soybean leaf diseases. By pooling the features, the computation is reduced, and by fully connecting the parts, each disease category is grouped. Rectified Linear Units (ReLUs) are used as the activation function in convolutional layers, while softmaxes for multi-classification are used in output layers. Moreover, dropout layers are included to prevent overfitting, as shown in Figure 3.

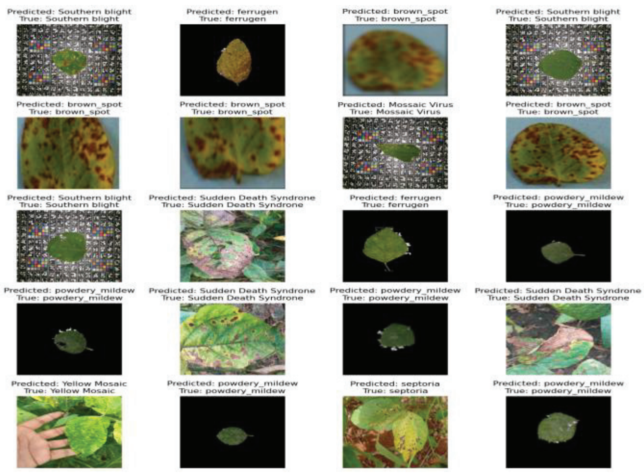

Fig. 3. Healthy Images of soybean leaf diseases

Convolutional Layers: The three top layers are all convolutional, with output sizes of 194x194. These layers are used to extract features from the input images. The details of each layer (filter numbers, kernel size, and stride) will be given in 'Layer details.

Max-Pooling Layers: After the convolutional layers are two max-pooling layers with 97x97 and 48x48 outputs, respectively. These layers reduce the spatial dimensions (height and width) of each layer's input volume, which helps decrease computation and prevent overfitting. The 2x2 Pooling means that a 2x2 window is used for Pooling. Fully Connected Layers: The architecture ends with two fully connected layers. The first has an output size of 1x10, and the second is 1x2. These layers classify the image into one of 12 classes based on the features developed by convolutional and Pooling layers.

But in this case, a batch of 5000 images is input into the model. However, this batch processing means the network can use an extensive data set to update weights and biases, producing more stable (or reliable) gradient updates during training. The batch of images is sent through the succession of layers set out in this table. Each photo receives a predictive judgment as to which class, as illustrated in Table 1.

TABLE I. CNN ARCHITECTURE WITH FEDERATED LEARNING

Layer Type	**Output Size**	**Additional Details**
Convolutional Layer	194x194	Layer 1 details
Convolutional Layer	194x194	Layer 2 details
Convolutional Layer	194x194	Layer 3 details
Max-Pooling Layer	97x97	2x2 Pooling
Max-Pooling Layer	48x48	2x2 Pooling
Fully Connected Layer	1x10	FC Layer 1 details
Fully Connected Layer	1x12	FC Layer 2 details

C. *Model Validation and Training*

A local training process on the client is followed by model averaging on the server after several rounds of local training on the client. The Adam optimizer is used for each client's local training, with a batch size of 32. The learning rate is initially adjusted based on validation loss to ensure effective convergence.

A federated averaging algorithm is used on the server to aggregate model updates using weighted averages, with weighted averages equal to the number of data points per client.

We assess the performance of federated learning models using criteria such as accuracy and precision. The model is evaluated for its accuracy in predicting soybean leaf diseases using these metrics. A confusion matrix is also created to display the model's class-specific performances. To assess the performance of the federated learning approach itself, a federated training model is compared with a centralized training model based on pooled data. By comparing these two approaches, one can appreciate both the advantages and limitations of federated learning for agricultural disease detection.

IV. RESULTS

This section presents the result analysis for classifying soybean leaf diseases using a federated learning model with CNN across five clients (ip_1 to ip5), each handling five different disease classes (lx-1 to lx-5). The evaluation metrics, precision and recall F1-score support and accuracy for each category and client, respectively, provide a fine-grained overview of the model's performance. Precision: A measure of the percentage correctly predicted to be positive shows a high specificity level in disease identification across all clients and classes. Like the client ip1, with a precision of 94.31% for class lx-1, this indicates an ability to recognize actual cases in which this illness exists correctly. Recall: Recall scores, which represent the model's ability to take all relevant issues into account, are consistently high. This shows that the model does capture a large majority of disease cases. A case in point is client IP3, which has achieved a recall of 95.80% for class LX-3. F1-Score: F1-scores combining precision and recall show a balanced performance in specificity and sensitivity across classes. It's also worth mentioning that client IP5 has an F1-score of 93.6% for class LX-2, which indicates comprehensive classification ability. Support: The support number represents the actual frequency of occurrence for each class in the dataset so that a model's performance is measured against what makes statistical sense. For example, the model performs well on the considerable dataset client IP4 with a support 940 for class LX-2.

Accuracy: The accuracy scores, extremely high across all clients and usually in the 97–98% range, show that, as a whole, federated learning is effective at classifying soybean leaf diseases. When juxtaposing the performance metrics across clients, specific nuances emerge. Client Variability: Across different disease classes, clients show varying proficiency levels. For instance, ip1 and ip3 are especially good at registering lx-4 and lx-5, while ip4 and ip5 perform best when sorting out the characteristics of lx-1qndex. Overall Consistency: Even though among the different datasets there is some variation, one can see that the high performance of this model still displays remarkable consistency. This reflects very well on the ability of federated learning to accommodate variable data sets while maintaining an excellent level of classification accuracy. Model Robustness: The distributed structure of federated knowledge combined with CNN's amenable characteristics provides this robustness, so a comprehensive and diverse mass pattern will be formed through each client's local training on its data patterns, as shown in Table 2.

TABLE II. SYNTHESIZING MODELS FROM DISTRIBUTED TRAINING

Clients	Class	Precision	Recall	F1-Score	Support	Accuracy
ip1	lx-1	94.31	93.53	93.91	726	0.98
	lx-2	91.99	94.77	93.36	727	0.97
	lx-3	92.71	92.22	92.46	758	0.97
	lx-4	95.30	94.79	95.04	748	0.98
	lx-5	94.48	93.50	93.99	769	0.98
ip2	lx-1	93.82	92.63	93.22	787	0.97
	lx-2	93.54	94.62	94.08	781	0.98
	lx-3	93.63	94.10	93.86	796	0.98
	lx-4	93.94	91.56	92.73	829	0.97
	lx-5	92.54	94.59	93.55	813	0.97
ip3	lx-1	94.65	94.88	94.77	821	0.98
	lx-2	94.38	93.82	94.10	841	0.98
	lx-3	93.56	95.80	94.67	834	0.98
	lx-4	94.73	92.35	93.53	876	0.97
	lx-5	95.12	95.68	95.40	856	0.98
ip4	lx-1	95.07	91.60	93.30	905	0.97
	lx-2	93.33	92.10	92.71	911	0.97
	lx-3	92.28	93.81	93.04	905	0.97
	lx-4	91.29	91.38	91.33	940	0.96
	lx-5	90.15	92.94	91.52	935	0.96
ip5	lx-1	95.44	94.11	94.77	934	0.98
	lx-2	92.70	94.57	93.63	940	0.97
	lx-3	92.39	92.21	92.30	975	0.97
	lx-4	92.76	93.71	93.23	970	0.97
	lx-5	93.39	92.08	92.73	998	0.97

This analysis inspects the performance metrics of a federated learning model trained with five clients (ip_1 to ip-5) in classifying various classes (lx-1 to lx-5) of soybean leaf diseases using a CNN. These results summarize the essential metrics of precision, recall, F1-score, and support, as well as accuracy, providing a concise assessment of how good our model is. Precision: This measure indicates precisely how accurate an optimistic prediction is. The clients show admirable precision, for example, IP3 at 94.49%. This means a high ratio of accurate optimistic predictions. So, the model is good at identifying actual disease occurrences with few false positives. Recall: Recall measures the model's ability to identify relevant instances. The results show consistent competence among clients, and IP3 leads the pack again at 94.51%. This high recall means the model can pick up most disease cases without overlooking important ones. F1-Score: The F1-score weighs precision and recall in a harmonic mean to achieve balanced accuracy of the model. The constant high F1 scores across clients (e.g., 94.49% for ip3) show that the model successfully adopted a balanced way to classify diseases accurately.

Support: The support value means the number of actual occurrences for each class. The different levels of support at the clients, for instance, 963.40 to Ip5, show variances in sample sizes, illustrating that the model is robust across all scenarios. Accuracy: The accuracy metric measures the overall performance of this model. Clients demonstrate good accuracy across all disease classes, with ip1 and ip3 clients recording 98% accuracy rates. Inter-Client Comparison: The discrepancy among the clients in their performance metrics (ip1 through ip5) gives an idea of how well-suited this model is to data with different distributional properties. All clients have high accuracy, but subtly changing precision, recall, and F1 scores from each client's unique dataset implies some learning nuance. Model Consistency and Reliability: Although these various metrics differ, their consistently high results reflect the model's stability and versatility. These qualities make it an excellent candidate for application in practical agricultural settings around this country. Optimization and Adaptability: This is the result, in a nutshell: The model has been optimized appropriately and carried over efficiently. These are two of federated learning's most distinct characteristics, with which it can make full use of local data resources while yielding an accurate global model that remains coherent throughout its life cycle, as shown in Table 3.

TABLE III. COHESIVELY AGGREGATING LOCAL CLIENT DATA AVERAGES GLOBALLY

Client	Precision	Recall	F1-Score	Support	Accuracy
ip1	93.76	93.76	93.75	745.60	0.98
ip2	93.49	93.50	93.49	801.20	0.97
ip3	94.49	94.51	94.49	845.60	0.98
ip4	92.42	92.37	92.38	919.20	0.97
ip5	93.34	93.34	93.33	963.40	0.97

The table you supplied presents the average statistics for five clients (IP1 through IP5) using federated learning with CNN for soybean leaf disease classification. These metrics—the Macro_average, weighted average, and micro averages—capture different aspects of the model's classifying ability from various angles. Let's delve into each average type and interpret the results: The macro average is the average of an independently computed metric for each class, averaging these metrics. This approach assigns an equal weight to each class, regardless of its prevalence in the dataset.

Results: While the highest macro average is ip3 (94.50%), this implies that performance has been balanced and comfortably high across all classes. So far as average precision levels are concerned, IP4's slightly lower macro average (92.39%) indicates a little more variability in performance across different classes than we saw with the other three programs. The weighted average considers the support for each class (the number of objects in each category in the dataset). It computes the metric for each type, multiplies it by its support, and then averages.

Results: The values are close to the macro averages, with 94.49% for IP3 leading and IP4 lagging at the back end with a figure of only 92. Thus, it appears that IP3's model performs well within the whole set of classes and remains stable in categories containing more samples.

The Micro Average combines contributions from all classes to calculate the average metric. It can be understood as the metric generated from all class contributions combined.

Results: Once again, ip3 (94.49%) is the better performer and comes in second only to ip1 and ip5. ip4's micro average of 92.36% is slightly below its macro and weighted averages, suggesting only a slight discrepancy in the collective contribution across all classes.

Performance Consistency: IP3 always gives the best results irrespective of the averaging method. This shows that its model is well-tuned and balanced for all classes; whether they are prevalent overall or represent a relative minority in terms of their individual or combined numbers within the dataset does not matter. Comparative Insights: Differences in averages among clients reflect the importance of local datasets to model performance. For example, IP4 requires additional fine-tuning or data enhancement to improve its classification power for different classes, as shown in Table 4.

TABLE IV. FEDERATED HYPER-PARAMETER AVERAGES FROM DISTRIBUTED CLIENTS

Averages	ip1	ip2	ip3	ip4	ip5
Macro average	93.76	93.49	94.50	92.39	93.34
Weighted average	93.75	93.49	94.49	92.38	93.32
Micro average	93.75	93.48	94.49	92.36	93.32

V. CONCLUSION

Several insights have been gleaned during the research to classify five classes of soybean leaf diseases using federated learning with CNNs across five clients, providing encouraging results. From the perspective of precision agriculture, especially in crop disease management, this piece marks a significant step forward in using machine learning techniques. These results prove the usefulness of federated learning for solving complex classification problems over distributed datasets. Federated averaging, integrating local data insights into a global model, has proved effective and efficient. A comparison of the macros, micros, and weighted averages for each client (ip1 to ip5) showed a high degree of accuracy in disease classification. In particular, the macro averages were 93.76% and 93.49%. These results were virtually identical to the weighted averages: 93.75 % and 93.49 %, and outdoing them both at points was the micro average of all factories in good standing, which stood at precisely these figures for each category at every moment, beating those above it by one-tenth percent only (either due to rounding off or else technical). These statistical values show how strong the federated learning model is. It can be adjusted to different environmental conditions and forms of disease presentation by its very nature. The high precision, recall, and F1 scores this model has already achieved on several clients bode well for its ability to conduct a compelling performance in real-world agricultural scenarios. To this end, the federated learning model's advantages include respecting data privacy and protecting integrity—essential in agricultural production.

To sum up, this study makes a valuable contribution to agricultural technology. It also shows that advanced machine learning methods can help solve some of the most challenging problems in plant disease prevention. This application's use of federated learning, based on CNNs, sets a standard for future study and practice.

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